Tables

Pose coverage and held-out image validation for multi-camera robot calibration

Table I. Complete-session counts and median outcomes across the four camera roles. Medians summarize role-specific fits and do not measure whole-robot accuracy. All fits use training observations only. Source: existing recordings and present calculations.

Session	Train poses	Held poses	Frames train/held	Park mm	Park px	Pixel fit px
A1	110	32	550/159	0.730	0.420	0.333
A2	110	32	549/160	0.526	0.419	0.375
B1	110	32	550/160	0.808	0.496	0.356
C1	110	32	550/160	0.658	0.422	0.314
D1	110	32	550/160	0.462	0.401	0.338
E1	110	32	550/160	0.755	0.491	0.349
F1	110	32	545/160	0.600	0.430	0.333
G1	110	32	550/160	0.642	0.410	0.343
H1	110	32	550/160	0.639	0.483	0.364

Table II. Main-cohort full-set solver outcomes. Numerical failure and successful-fit medians retain their denominators. Different successful subsets prevent a universal solver ranking. Source: present calculations.

Solver	Finite / attempted	Failed	Median closure mm	Median held-image px
Tsai	9/36	27	5.050	2.555
Park	36/36	0	0.647	0.443
Horaud	36/36	0	0.647	0.443
Andreff	27/36	9	1.474	0.624
Daniilidis	36/36	0	0.632	0.436

Table III. Ten-pose farthest-point/random comparison. Ratios use each panel’s median random error; archive summaries first geometrically average panels within a robot. Values below one favour farthest-point selection. Source: present calculations.

Cohort	Closure < 1	Median ratio	Image < 1	Median ratio	Archive ratios closure/image
Original	9/12	0.833	8/12	0.920	0.882/0.941
Extension	17/24	0.879	17/24	0.879	0.883/0.919
All main	26/36	0.854	25/36	0.912	0.883/0.924
Ancillary	20/23	0.885	19/23	0.885	0.892/0.915
ID sensitivity	23/32	0.854	22/32	0.912	0.904/0.927